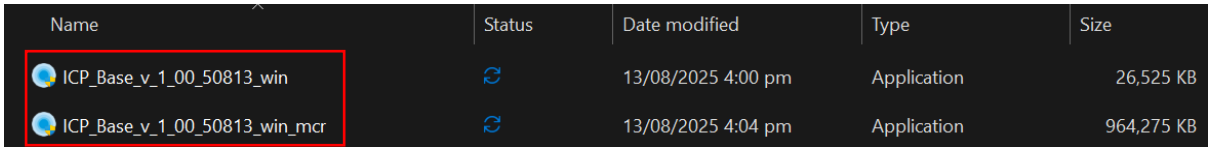
1. How to get ICP-Base running

1.1. Compiled version

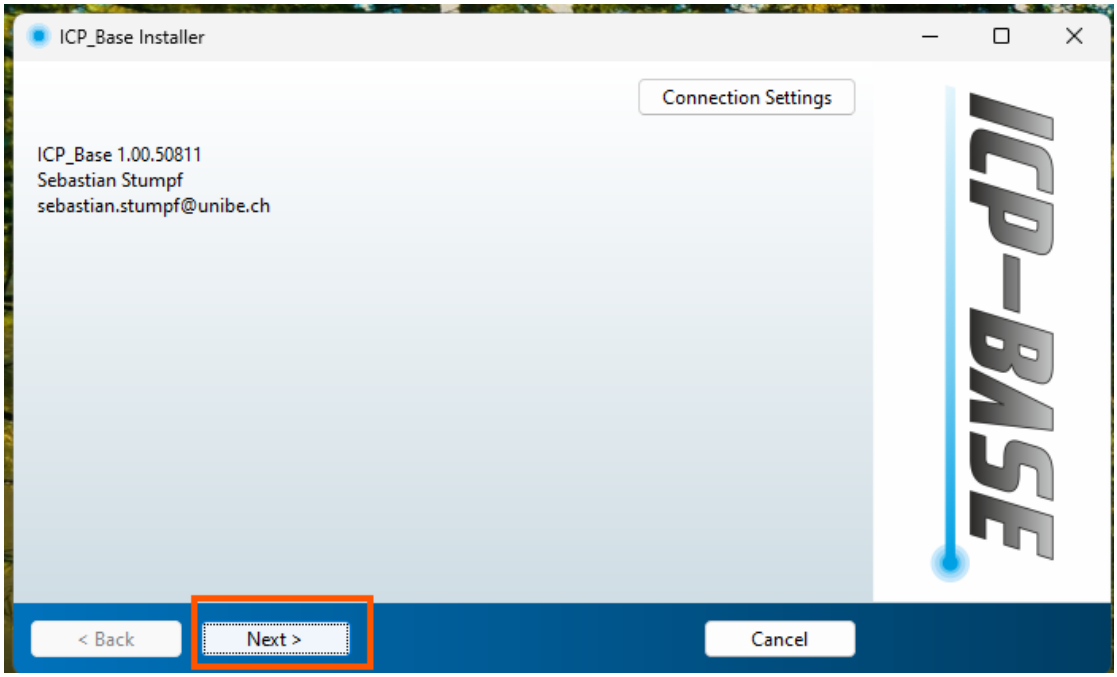
(a) If the filename of the install-file (.exe) is NOT ending in "_mcr" you need internet access to download the MATLAB Runtime library. So you need internet access to run ICP-Base.

(b) If the filename is ending with "_mcr" you do not need internet access since the Runtime Library is already included and you can install ICP-Base on any computer, even without internet access. This file is therefore also much larger in memory.

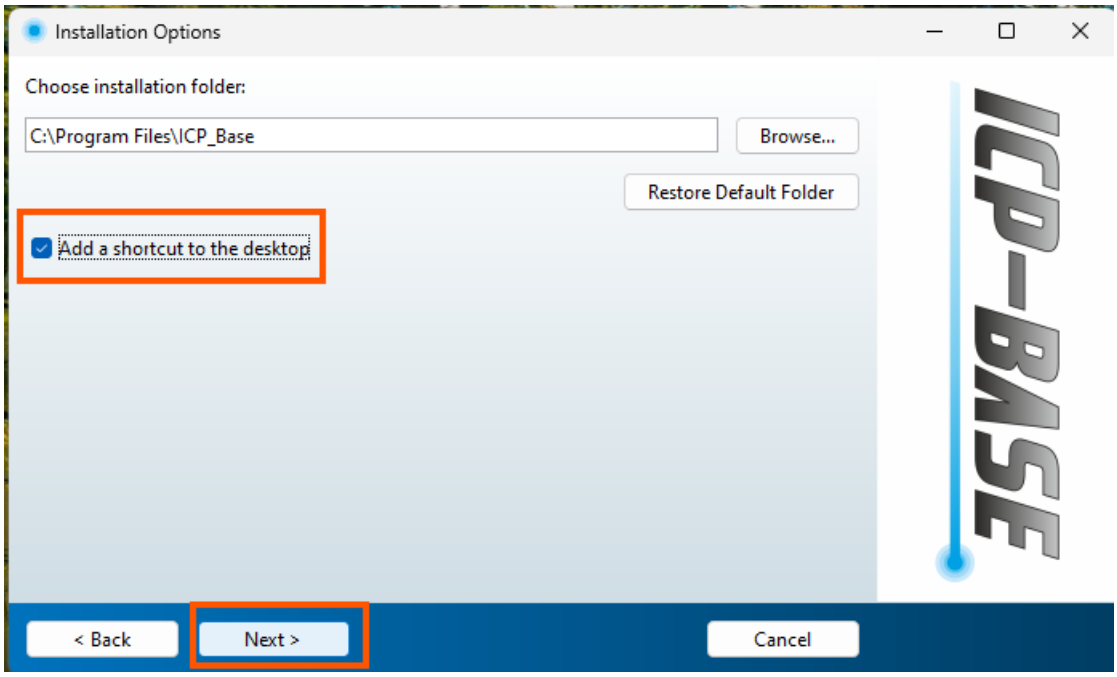


Name	Status	Date modified	Type	Size
ICP_Base_v_1_00_50813_win		13/08/2025 4:00 pm	Application	26,525 KB
ICP_Base_v_1_00_50813_win_mcr		13/08/2025 4:04 pm	Application	964,275 KB

Simply double-click the install-file (e.g. ICP_Base_v_1_XX_XX.exe) and the installation process starts. This is shown for windows but is analogous for MacOS.

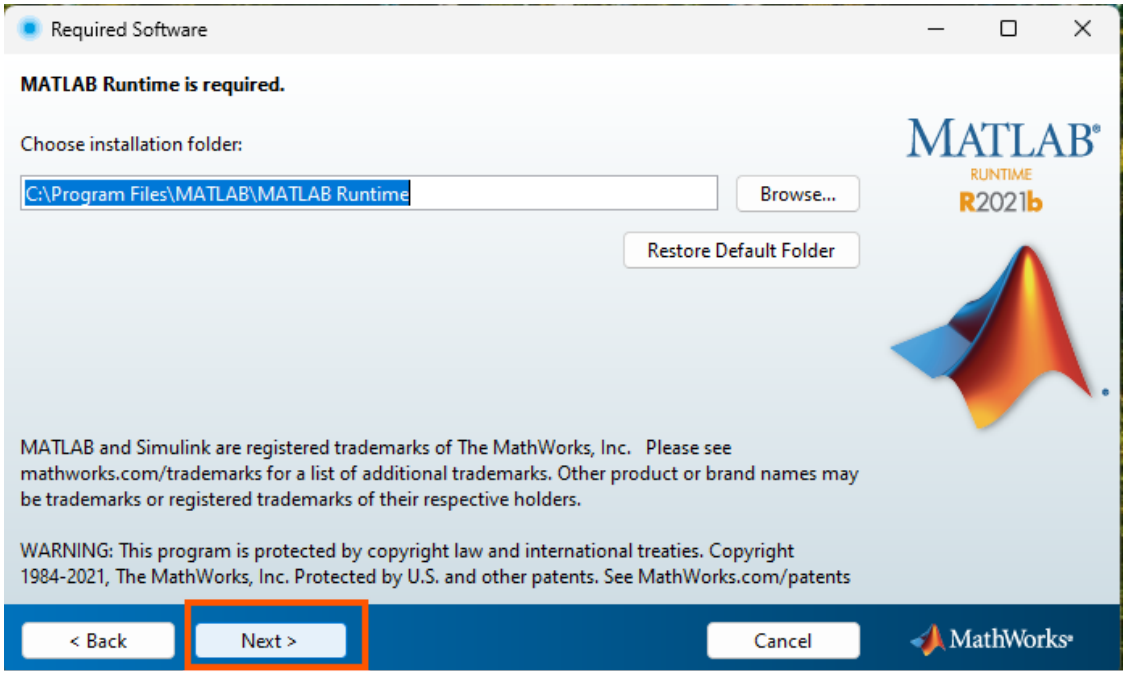


Click "Next >".

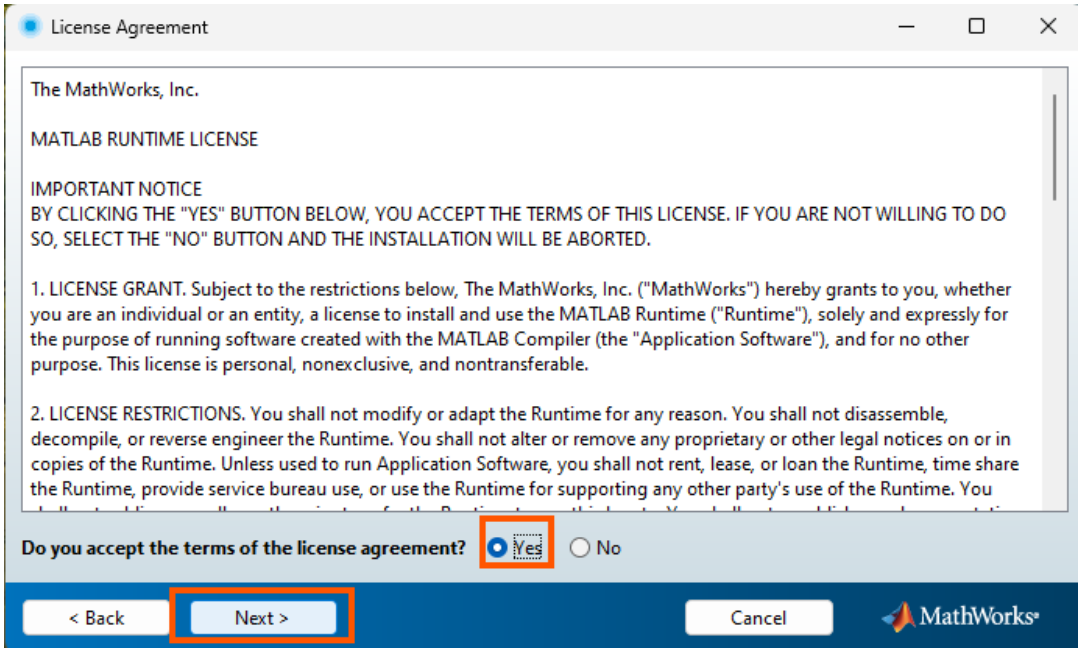


I would recommend to add a shortcut to the desktop. **Also make sure to remember where ICP-Base installation folder is located because you might want to access it later to customize your plots and reference material files!**

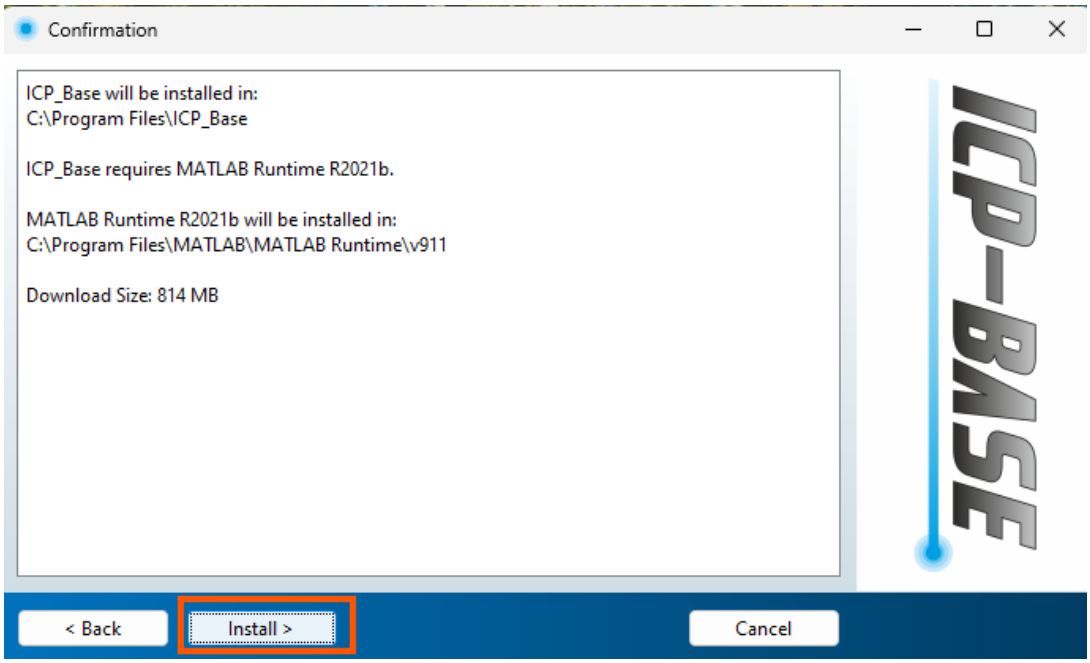
Then click “Next >”.



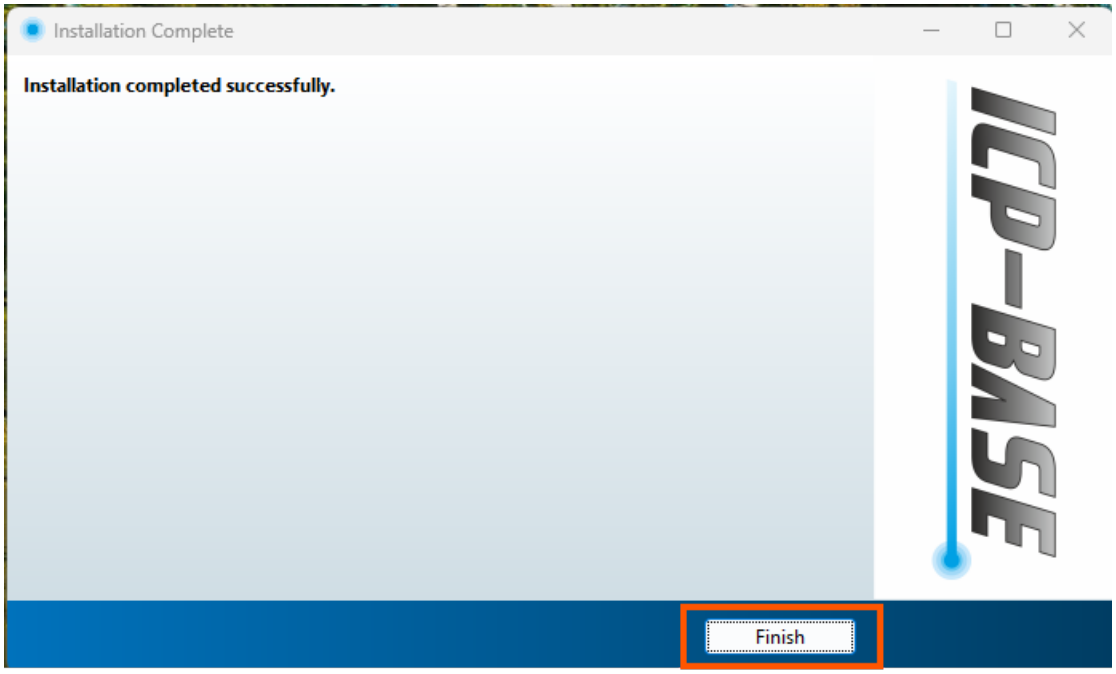
If you do not have the Runtime library already installed, MATLAB will do that for you. Just click “Next >”.



Agree and click “Next >”.

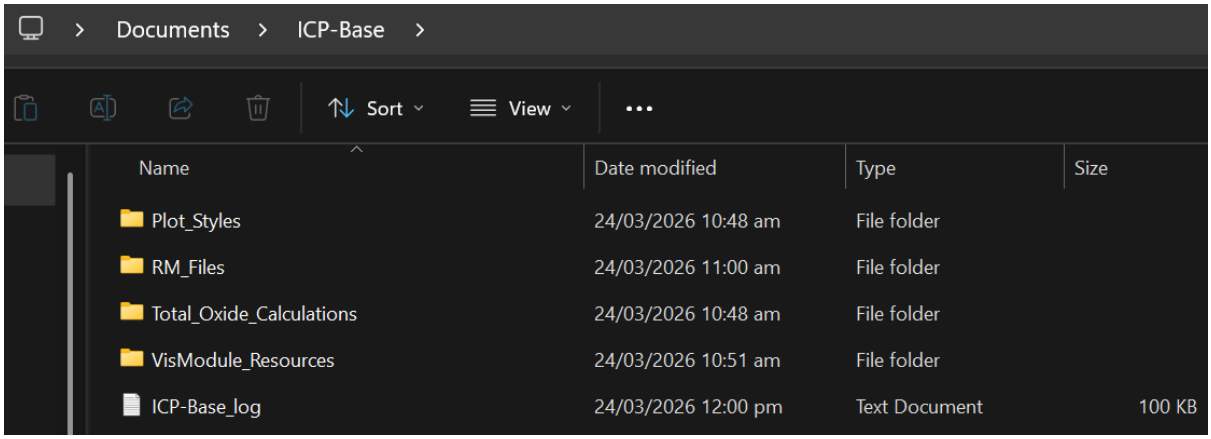


Click “Install”.



You did it! Click “Finish” and the shortcut icon (if selected above) should show up on your desktop. Now start ICP-Base by double-clicking the shortcut icon. This can take some time since MATLAB initializes the graphical user interface for the first time.

In the path Documents/ICP_Base, you will find subfolders including “RM_Files” and “Plot_Styles” and “Total_Oxide_Calculations” which can be important depending on your use case.

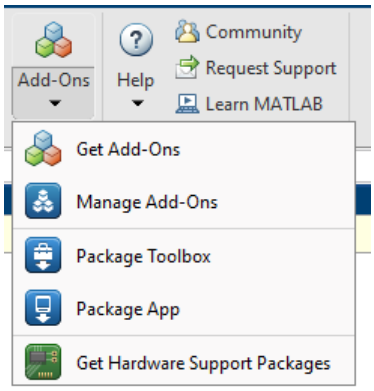


1.2. Uncompiled version

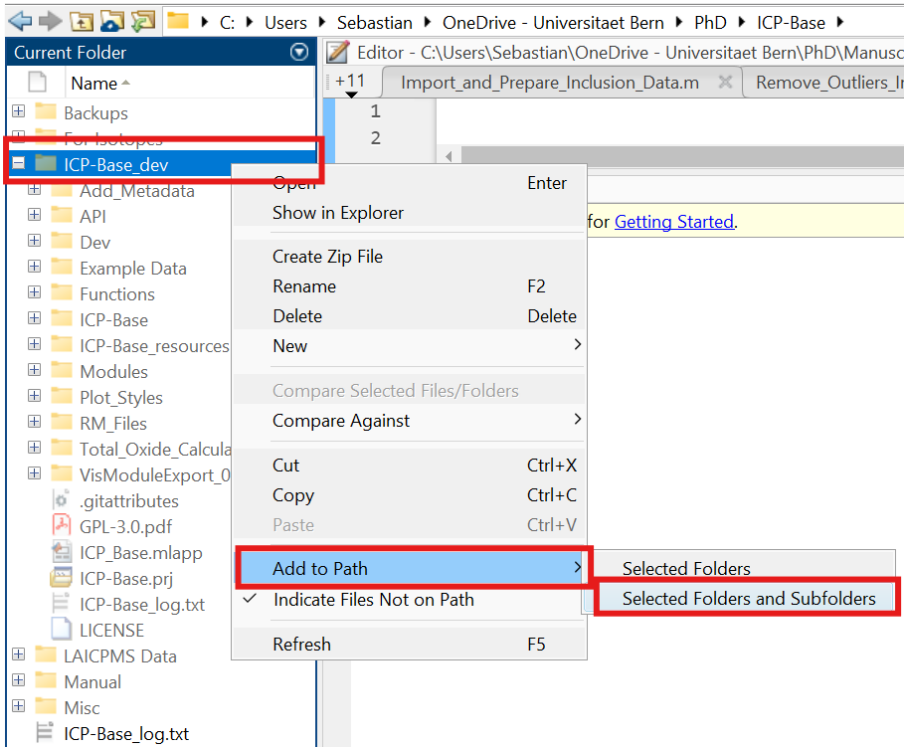
If you choose to run it uncompiled, make sure to have MATLAB and the MATLAB App Designer with a valid license. Ideally you use the **MATLAB 2021b** version since this is the same version the software was programmed in initially. Bugs can occur if you use a newer version.

When installing MATLAB make sure to add the Add-Ons "Image Processing Toolbox" and "Statistics and Machine Learning Toolbox".

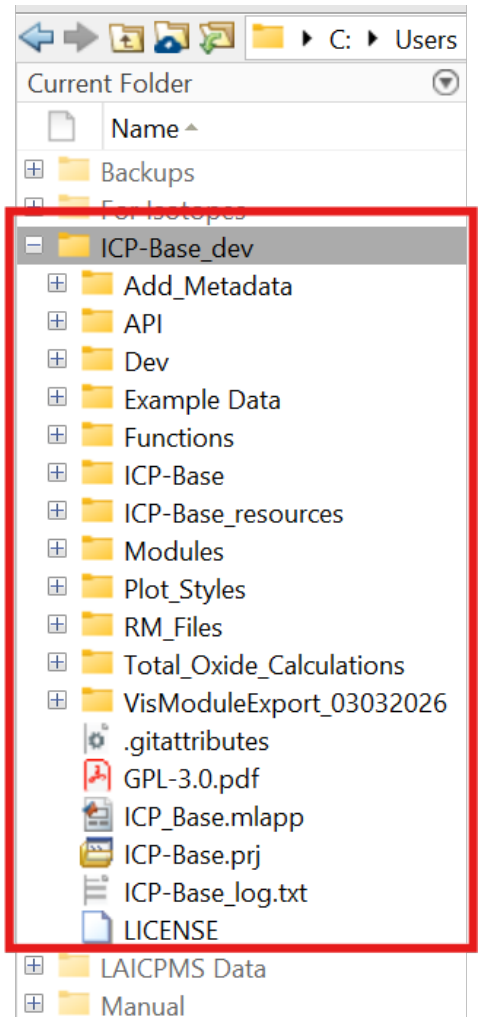
If your MATLAB is already set up but those Add-Ons are not installed you can get them using the "Add-Ons" Button in the "Home" Tab.



Once your MATLAB is running, navigate the path to the folder in which ICP-Base is stored. Then right-click the folder and selected "Add to Path" followed by "Select Folders and Subfolders".



The folders are then not greyed out anymore:



Next, either right-click on ICP_Base.mlapp and press “Run” to open it directly or double-click on it to open ICP-Base in the App Designer. Once ICP-Base is open in the App Designer, go to the “Editor” or “Designer” Tab and then press the “Run” button.